

Experiment 4a: Comparative Analysis of Large Language Models

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Aim:

To develop a Python program that performs comparative analysis of Large Language Models by analyzing text-generation performance.

Models Used:

- GPT2
- DistilGPT2
- OPT-350M (Hugging Face Models)

Common Prompt Used:

Write a haiku about AI and nature.

Program Code:

```
import gradio as gr
from transformers import pipeline

model1 = pipeline("text-generation", model="gpt2") model2 =
pipeline("text-generation", model="distilgpt2")
model3 = pipeline("text-generation", model="facebook/opt-350m")

def compare_models(prompt):
    output1 = model1(prompt, max_length=50)[0]["generated_text"]
    output2 = model2(prompt, max_length=50)[0]["generated_text"]
    output3 = model3(prompt, max_length=50)[0]["generated_text"]

    scores = {
        "GPT2": len(output1),
        "DistilGPT2": len(output2), "OPT-
        350M": len(output3)
    }

    ranked = sorted(scores.items(), key=lambda x: x[1], reverse=True) ranking_text =
    "\n".join([f"{i+1}. {m}" for i, (m, _) in enumerate(ranked)])

    return output1, output2, output3, ranking_text

interface = gr.Interface( fn=compare_models,
    inputs=gr.Textbox(label="Enter Prompt"), outputs=[
        gr.Textbox(label="GPT2 Output"),
        gr.Textbox(label="DistilGPT2 Output"),
```

```
        gr.Textbox(label="OPT-350M Output"),
        gr.Textbox(label="Ranking (Best → Worst) ")

    ],
    title="Comparative Analysis of LLMs"
)

interface.launch()
```

Result:

The comparative analysis of three Large Language Models was successfully performed using Hugging Face. The outputs were generated for a common prompt and ranked based on evaluation criteria.

Learning Outcome:

Students were able to deploy and compare multiple LLMs, evaluate text generation performance, and rank models based on output quality and relevance.